

CMOS VLSI Design Project

Design and Analysis of a 2:1 CMOS Multiplexer

Using Cadence Virtuoso

Submitted By:

Name: Piyal Chakraborty

Student ID: 2108028

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Department of Electronics and Telecommunication Engineering, CUET

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Contents

1	Introduction	2
2	Objectives	2
3	Theory of CMOS Multiplexer	3
4	Transistor Sizing	4
5	Circuit Design	4
5.1	Schematic Design	4
5.2	Multiplexer Symbol	5
6	Physical Layout Design	6
7	Design Rule Check (DRC)	7
8	Layout Versus Schematic (LVS)	7
9	RCX and Parasitic Extraction	8
10	Post-Layout Simulation	9
11	Average Power Calculation	9
12	Propagation Delay Calculation	10
13	Verification and Performance Summary	11
14	Discussion	12
15	Conclusion	13
16	Software Used	14

1 Introduction

The multiplexer is an important building block of digital integrated circuits. It is used to select one of multiple input signals and transfer the selected signal to a single output.

In this project, a two-to-one CMOS multiplexer was designed and analyzed using the Cadence Virtuoso design environment. The complete custom CMOS design flow included schematic design, symbol generation, physical layout design, Design Rule Check (DRC), Layout Versus Schematic (LVS) verification, RC parasitic extraction (RCX), and post-layout simulation.

The post-layout simulation was used to evaluate important performance parameters such as average power consumption and propagation delay.

A two-to-one multiplexer has two data inputs, A and B , one select input S , and one output Y .

The Boolean expression of a 2:1 multiplexer is

$$Y = \overline{S}A + SB \quad (1)$$

where A and B are the data inputs, S is the select signal, and Y is the output.

2 Objectives

The main objectives of this project are:

- To design a two-to-one CMOS multiplexer using Cadence Virtuoso.
- To create and verify the transistor-level schematic of the multiplexer.
- To generate a reusable symbol for the designed multiplexer.
- To design the physical layout of the CMOS multiplexer.
- To verify the physical layout using Design Rule Check (DRC).
- To verify the electrical equivalence between the schematic and layout using Layout Versus Schematic (LVS).
- To perform parasitic resistance and capacitance extraction using RCX.
- To perform post-layout transient simulation using the extracted circuit.
- To calculate the average power consumption of the multiplexer.
- To calculate the propagation delay of the multiplexer.

3 Theory of CMOS Multiplexer

A two-to-one multiplexer is a combinational digital circuit that selects one of two input signals according to the value of a select signal.

The two data inputs are denoted by A and B , while the select input is denoted by S . The output is represented by Y .

The Boolean expression of the multiplexer is

$$Y = \bar{S}A + SB. \quad (2)$$

When the select signal is LOW, $S = 0$, the output becomes

$$Y = \bar{0}A + 0B \quad (3)$$

and therefore

$$Y = A. \quad (4)$$

Thus, when $S = 0$, input A is selected.

When the select signal is HIGH, $S = 1$, the output becomes

$$Y = \bar{1}A + 1B \quad (5)$$

and therefore

$$Y = B. \quad (6)$$

Thus, when $S = 1$, input B is selected.

The truth table of the two-to-one CMOS multiplexer is shown in Table 1.

Table 1: Truth Table of a Two-to-One CMOS Multiplexer

S	A	B	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

The operation of the multiplexer can therefore be summarized as

$$Y = \begin{cases} A, & S = 0, \\ B, & S = 1. \end{cases} \quad (7)$$

4 Transistor Sizing

The CMOS multiplexer was designed using the required PMOS and NMOS transistors according to the transistor-level implementation in Cadence Virtuoso.

The transistor dimensions used in the design are summarized below.

Table 2: Transistor Dimensions Used in the CMOS Multiplexer

Transistor	Width	Length
PMOS	480 nm	100 nm
NMOS	240 nm	100 nm

Therefore, the PMOS-to-NMOS width ratio used in the design is

$$\frac{W_p}{W_n} = \frac{480}{240} = 2. \quad (8)$$

The PMOS transistors were designed with approximately twice the width of the NMOS transistors. This larger PMOS width helps compensate for the lower carrier mobility of holes compared with electrons and provides adequate pull-up strength.

5 Circuit Design

5.1 Schematic Design

The two-to-one CMOS multiplexer schematic was created using the Cadence Virtuoso Schematic Editor.

The circuit contains two data input terminals, A and B , one select terminal, S , and one output terminal, Y .

The transistor-level implementation was designed to realize the Boolean function

$$Y = \bar{S}A + SB. \quad (9)$$

The schematic of the designed multiplexer is shown in Figure 1.

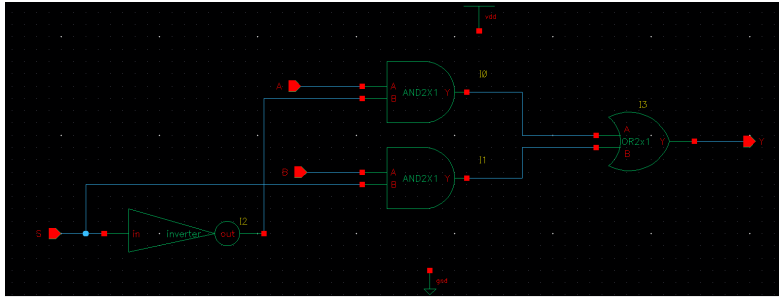


Figure 1: Schematic of the two-to-one CMOS multiplexer in Cadence Virtuoso.

The logical operation of the circuit can be represented as

$$A, B, S \rightarrow \bar{S}A + SB. \quad (10)$$

5.2 Multiplexer Symbol

After completing the schematic, a symbol was generated for the two-to-one CMOS multiplexer.

The symbol contains two data input terminals labeled A and B , one select terminal labeled S , and one output terminal labeled Y .

The generated multiplexer symbol is shown in Figure 2.

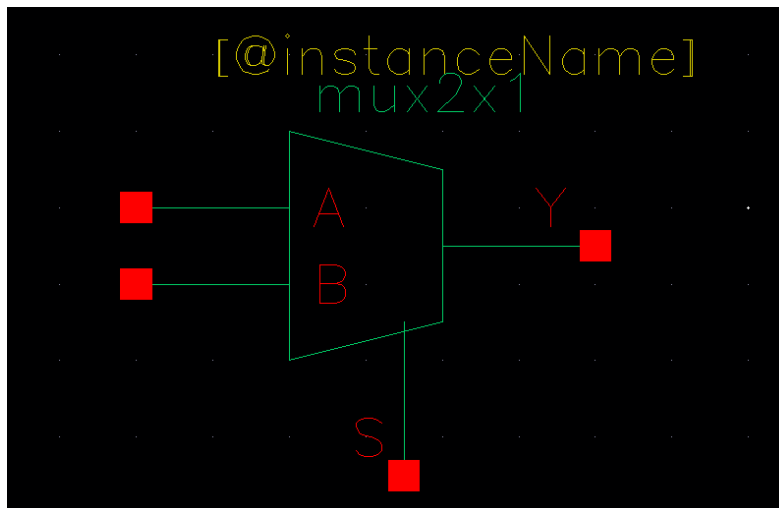


Figure 2: Generated symbol of the two-to-one CMOS multiplexer.

The symbol allows the designed multiplexer to be used conveniently as a reusable hierarchical block in larger digital circuits.

6 Physical Layout Design

After completing the schematic and symbol generation, the physical layout of the CMOS multiplexer was developed using Cadence Virtuoso Layout Editor.

The layout represents the physical implementation of the transistor devices and their electrical interconnections.

The physical layout contains the required layers for transistor formation and electrical interconnection, including:

- active or diffusion regions,
- polysilicon,
- N-well,
- metal layers,
- contacts,
- vias,
- power connection, and
- ground connection.

The data inputs and select signal are routed to the appropriate transistor gates and the transistor networks are interconnected to produce the selected output.

The completed physical layout is shown in Figure 3.

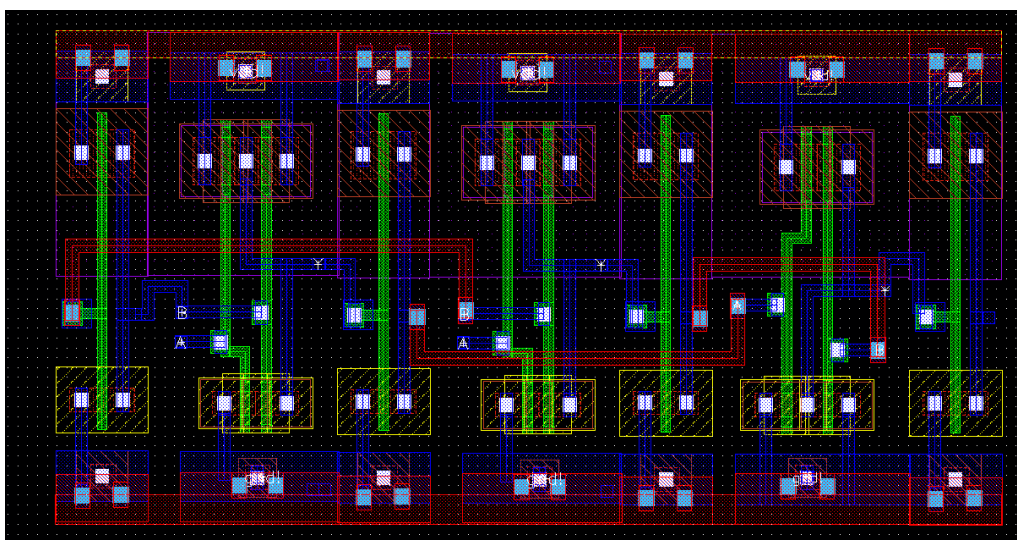


Figure 3: Physical layout of the two-to-one CMOS multiplexer.

The layout was designed while maintaining the required minimum width, spacing, enclosure, and connectivity rules of the selected CMOS technology.

7 Design Rule Check (DRC)

Design Rule Check (DRC) is performed to verify whether the physical layout satisfies the fabrication rules of the selected CMOS technology.

DRC checks various geometrical constraints such as:

- minimum metal width,
- minimum metal spacing,
- minimum polysilicon width,
- polysilicon spacing,
- diffusion spacing,
- contact dimensions,
- contact enclosure,
- via enclosure, and
- well spacing.

The completed CMOS multiplexer layout was checked using the Cadence DRC verification flow.

DRC Result: PASSED

The physical layout satisfies the required design rules of the selected CMOS technology.

Thus, the physical layout is valid with respect to the required fabrication design rules.

8 Layout Versus Schematic (LVS)

After successful DRC verification, Layout Versus Schematic (LVS) verification was performed.

LVS compares the electrical netlist extracted from the physical layout with the netlist generated from the original schematic.

It verifies important circuit properties such as:

- number of MOS transistors,
- PMOS and NMOS device types,
- gate connectivity,

- source and drain connectivity,
- data input connections,
- select input connection,
- output connection,
- power connection, and
- ground connection.

The final multiplexer layout was checked against the original schematic.

LVS Result: PASSED

The physical layout is electrically equivalent to the original schematic.

The successful LVS result confirms that the intended electrical connectivity was maintained during the physical layout design.

9 RCX and Parasitic Extraction

After DRC and LVS verification, RC parasitic extraction was performed using the RCX extraction flow.

Physical interconnections in an integrated circuit introduce additional parasitic resistance and capacitance.

These parasitic elements can affect important performance parameters such as propagation delay and power consumption.

RCX extracts these parasitic resistance and capacitance components from the physical layout.

The extracted view of the designed multiplexer is shown in Figure 4.

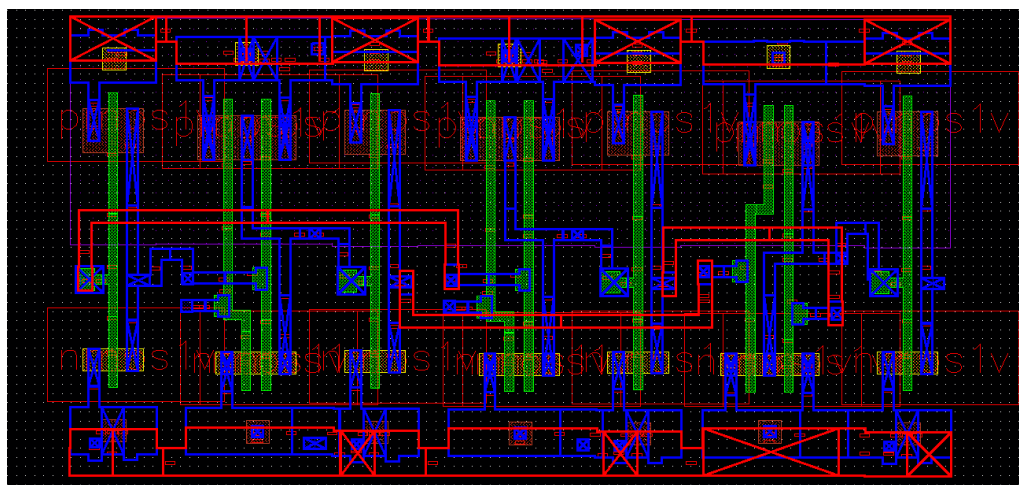


Figure 4: RCX/extracted view of the two-to-one CMOS multiplexer.

RCX: Successfully Completed

The extracted multiplexer circuit was used for post-layout simulation.

10 Post-Layout Simulation

After parasitic extraction, post-layout transient simulation was performed using the RC-extracted view of the CMOS multiplexer.

Unlike ideal schematic-level simulation, post-layout simulation includes the effects of parasitic resistance and capacitance generated by the physical layout.

Therefore, post-layout simulation provides a more realistic estimation of the circuit performance.

The main performance parameters calculated in this project were:

1. Average power consumption
2. Propagation delay

11 Average Power Calculation

The average power consumption of the CMOS multiplexer was determined from the post-layout transient simulation results using the Cadence Virtuoso Calculator.

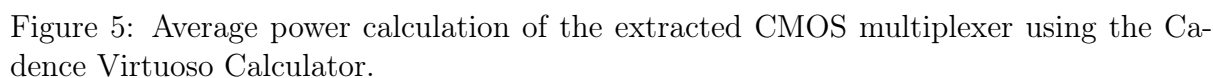
The average power over a time interval T can be expressed as

$$P_{\text{avg}} = \frac{1}{T} \int_0^T p(t) dt \quad (11)$$

where $p(t)$ represents the instantaneous power consumption.

The average value of the simulated power waveform was calculated using the Cadence Virtuoso Calculator.

The power calculation is shown in Figure 5.


$$P_{\text{avg}} = 10.65 \times 10^{-6} \text{ W.} \quad (12)$$
$$1 \mu\text{W} = 10^{-6} \text{ W}, \quad (13)$$
$$P_{\text{avg}} = 10.65 \mu\text{W} \quad (14)$$
$$\boxed{10.65 \mu\text{W}}. \quad (15)$$

12 Propagation Delay Calculation

For the multiplexer, a selected input transition that causes the output to change was used to determine the propagation delay.

$$t_p = t_{\text{out}} - t_{\text{in}}. \quad (16)$$

10

The Cadence Virtuoso Calculator was used to determine the delay between the selected input transition and the corresponding output transition.

The delay calculation is shown in Figure 6.

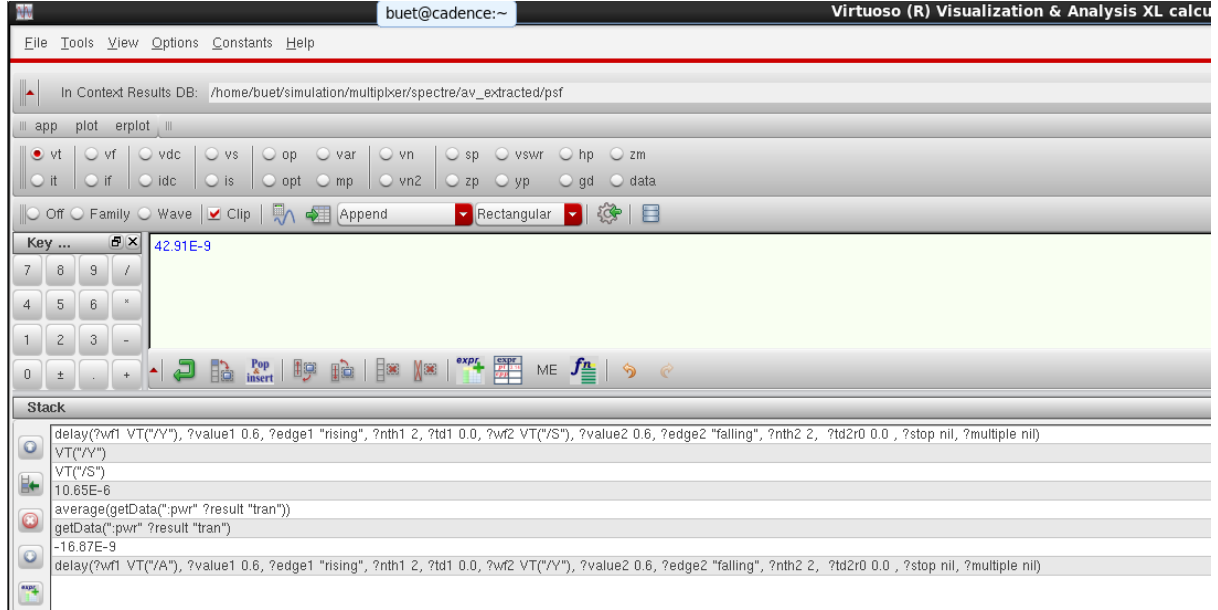


Figure 6: Propagation-delay calculation of the extracted CMOS multiplexer using the Cadence Virtuoso Calculator.

The measured propagation delay obtained from the Cadence Calculator is

$$t_p = 42.91 \times 10^{-9} \text{ s.} \quad (17)$$

Since

$$1 \text{ ns} = 10^{-9} \text{ s,} \quad (18)$$

the propagation delay becomes

$$t_p = 42.91 \text{ ns} \quad (19)$$

Therefore, the measured post-layout propagation delay for the selected input-output transition of the CMOS multiplexer is approximately

$$42.91 \text{ ns.} \quad (20)$$

13 Verification and Performance Summary

The overall design, verification, and performance results of the two-to-one CMOS multiplexer are summarized in Table 3.

Table 3: Summary of the CMOS Multiplexer Design Results

Parameter	Result
Logic Function	$Y = \bar{S}A + SB$
Transistor Dimensions	$W_p = 480 \text{ nm}$, $W_n = 240 \text{ nm}$, $L = 100 \text{ nm}$
Schematic Design	Completed
Symbol Generation	Completed
Physical Layout	Completed
DRC	Passed
LVS	Passed
RCX / Parasitic Extraction	Completed successfully
Post-layout Simulation	Completed
Average Power	$10.65 \mu\text{W}$
Propagation Delay	42.91 ns

14 Discussion

The complete custom CMOS design flow for a two-to-one multiplexer was carried out using Cadence Virtuoso.

The multiplexer performs the selection of one of two data inputs based on the value of the select signal. The required Boolean function is

$$Y = \bar{S}A + SB. \quad (21)$$

When the select signal is LOW, the output follows input A . When the select signal is HIGH, the output follows input B .

After completing the transistor-level schematic, a reusable multiplexer symbol was generated.

The physical layout was then developed using the required active, polysilicon, well, metal, contact, via, power, and ground layers.

The layout was checked using Design Rule Check and Layout Versus Schematic verification.

After successful physical verification, RC parasitic extraction was performed using the RCX flow.

The extracted circuit was then used for post-layout transient simulation so that the effects of layout-dependent parasitic resistance and capacitance could be included in the

performance analysis.

The average post-layout power consumption obtained from the Cadence Virtuoso Calculator was approximately

$$P_{\text{avg}} = 10.65 \mu\text{W}. \quad (22)$$

The measured propagation delay for the selected input-output transition was approximately

$$t_p = 42.91 \text{ ns}. \quad (23)$$

These results include the effects of layout-dependent parasitic elements and therefore provide a more realistic indication of the circuit performance than an ideal schematic-level simulation.

15 Conclusion

A two-to-one CMOS multiplexer was successfully designed, implemented, and analyzed using Cadence Virtuoso.

The required Boolean function was

$$Y = \overline{S}A + SB. \quad (24)$$

The multiplexer selects input A when the select signal S is LOW and selects input B when the select signal S is HIGH.

The project covered the major stages of the custom CMOS VLSI design flow, including schematic design, symbol generation, physical layout design, DRC verification, LVS verification, RC parasitic extraction, and post-layout simulation.

The physical layout was checked against the required fabrication design rules and its electrical connectivity was verified against the original schematic.

RCX was also completed successfully, allowing parasitic resistance and capacitance to be included in the final simulation.

The obtained post-layout average power consumption was

$$\boxed{P_{\text{avg}} = 10.65 \mu\text{W}} \quad (25)$$

and the measured propagation delay was

$$\boxed{t_p = 42.91 \text{ ns}.} \quad (26)$$

Therefore, the project demonstrates the complete schematic-to-layout design, verification, extraction, and performance analysis of a two-to-one CMOS multiplexer using Cadence Virtuoso.

16 Software Used

The following software tools were used to complete the project:

- Cadence Virtuoso Schematic Editor
- Cadence Virtuoso Layout Editor
- Cadence Spectre Simulator
- Design Rule Check (DRC) Tool
- Layout Versus Schematic (LVS) Tool
- RCX / Parasitic Extraction Tool
- Cadence Virtuoso Visualization and Analysis Calculator